

4.13.27

EE25BTECH11022 - sankeerthan

Question: Consider the set of all lines $px + qy + r = 0$ such that $3p + 2q + 4r = 0$. Which one of the following statements is true?

- 1) The lines are concurrent at the point $(\frac{3}{4}, \frac{1}{2})$.
- 2) Each line passes through the origin.
- 3) The lines are all parallel.
- 4) The lines are not concurrent.

Solution: Given the set of lines $px + qy + r = 0$ such that $3p + 2q + 4r = 0$.

$$\begin{pmatrix} x & y & 1 \\ 3 & 2 & 4 \end{pmatrix} \begin{pmatrix} p \\ q \\ r \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad (1)$$

For a non-trivial solution, the vector (p, q, r) must be in the nullspace. The system has a non-trivial solution if and only if the row vectors are linearly dependent:

$$(x_0, y_0, 1) = \lambda (3, 2, 4) \quad (2)$$

$$x_0 = 3\lambda, y_0 = 2\lambda, 1 = 4\lambda \quad (3)$$

$$1 = 4\lambda \implies \lambda = \frac{1}{4} \quad (4)$$

$$x_0 = \frac{3}{4}, \quad y_0 = \frac{1}{2} \quad (5)$$

Therefore, all lines in the given family are concurrent at the point $(\frac{3}{4}, \frac{1}{2})$

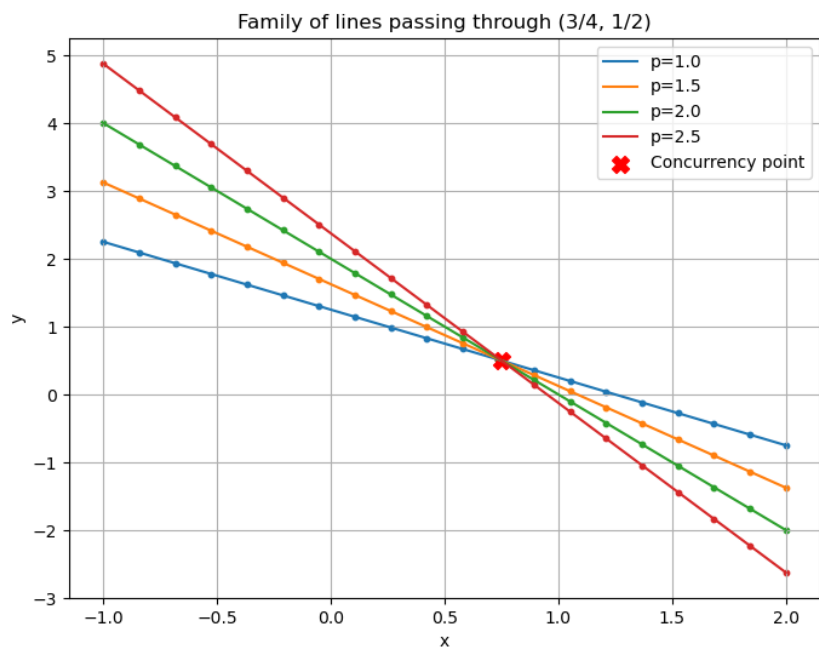


Fig. 4.1